

Cloud Computing Applications and Services

(Aplicações e Serviços de Computação em Nuvem)

Guide 1: Swap

University of Minho

2025-2026

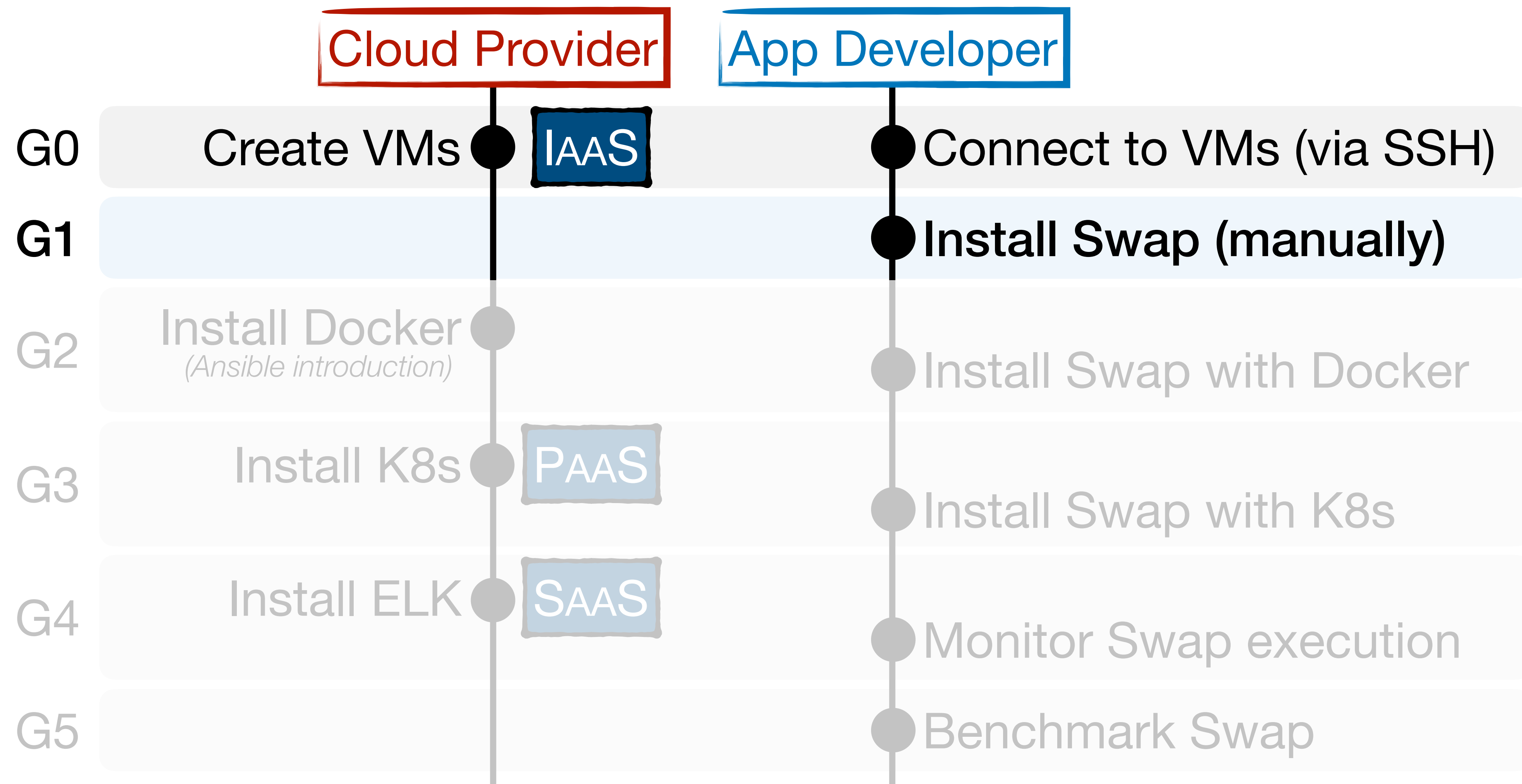


Context

- With “Guide 0”, you (as a *cloud provider*) have set up several VMs for potential cloud users.
- Now, how can an application developer use these VMs to install and deploy applications?
- The goal of this guide (as an *application developer*) is to install and configure a distributed application (**Swap**) on these VMs.

Road Map

Where are you on the roadmap?

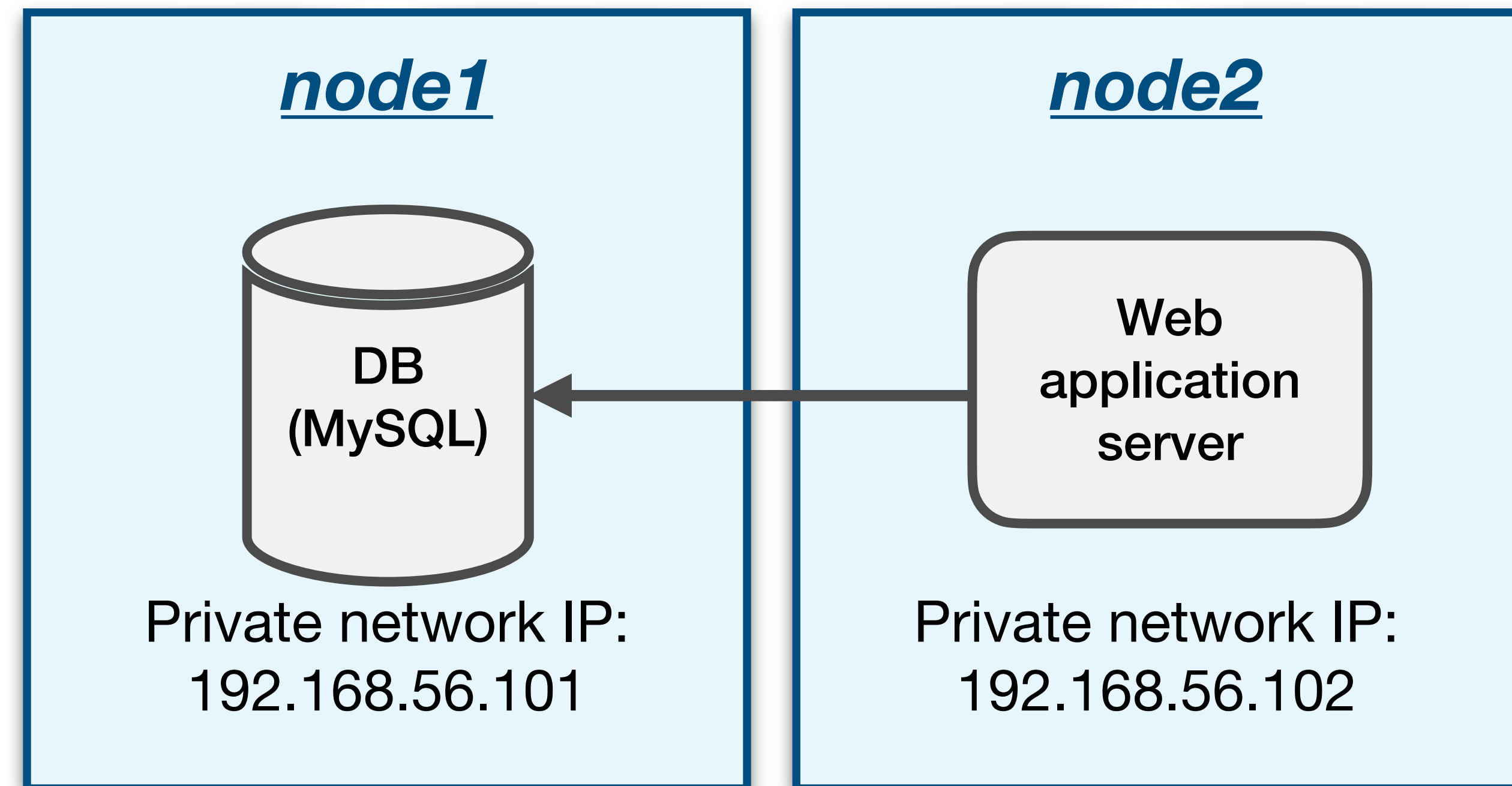


Goal

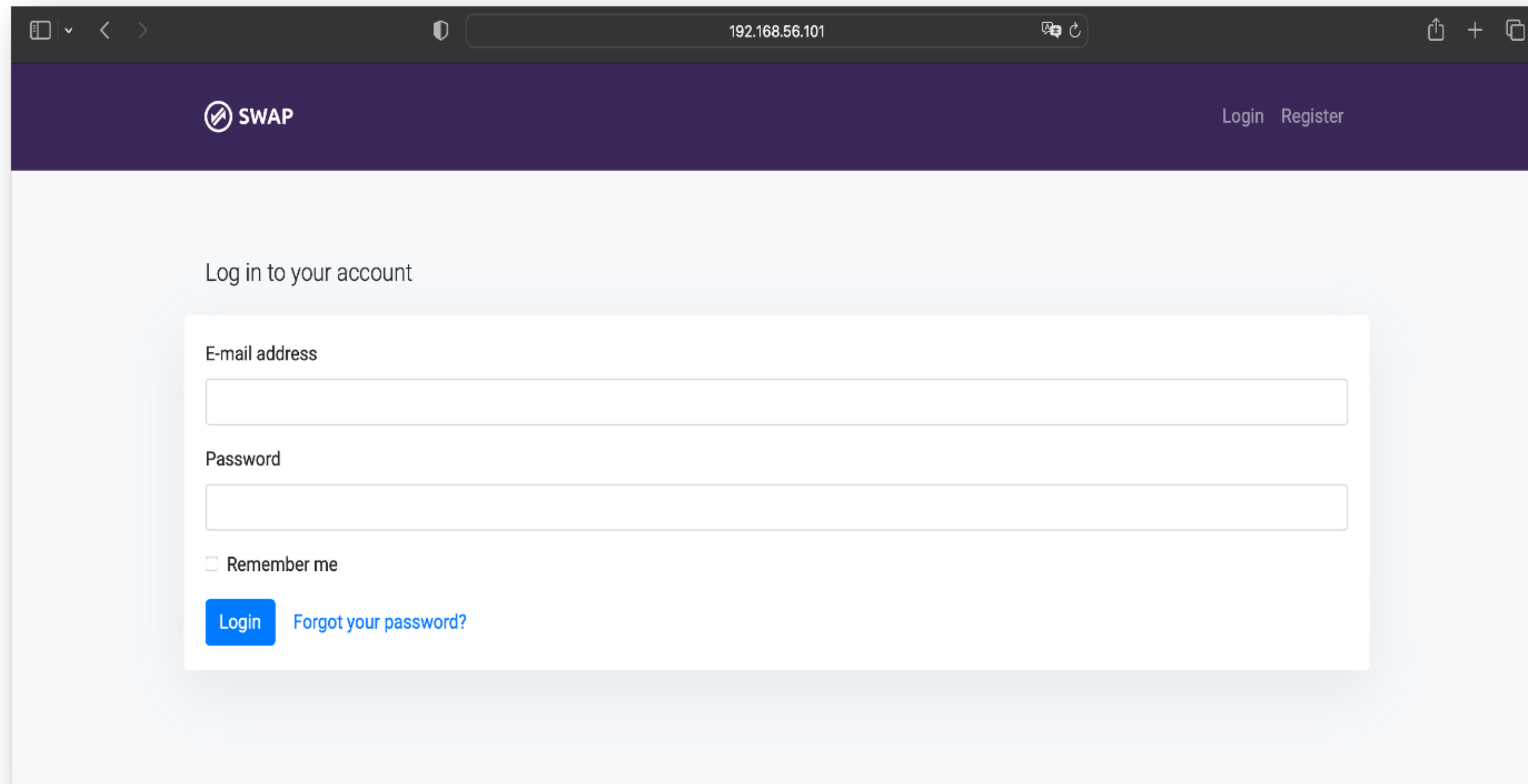
App Developer

● Install the Swap application

- Swap's database will run on *node1*
- Swap's web application server will run on *node2*



Swap: Basic Usage



The screenshot shows a web browser window with the address bar displaying '192.168.56.101'. The page has a dark purple header with the 'SWAP' logo on the left and 'Login Register' links on the right. The main content area is light blue and contains the text 'Log in to your account'. Below this is a white login form with two input fields: 'E-mail address' and 'Password'. There is a 'Remember me' checkbox and a blue 'Login' button. A link for 'Forgot your password?' is also present.

Log in to your account

E-mail address

Password

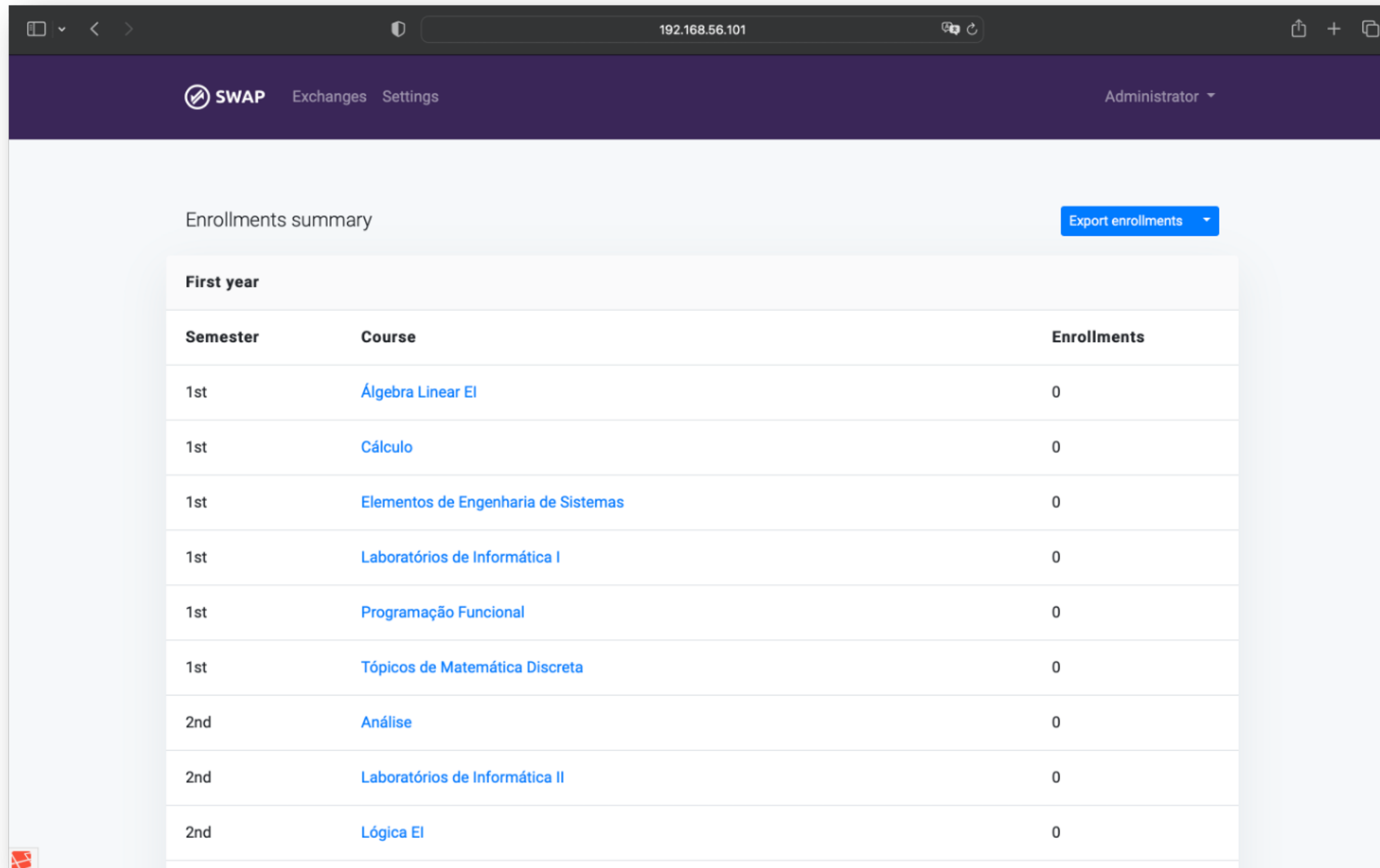
☐ Remember me

Login [Forgot your password?](#)

Email address: contact@hackathonners.org

Password: "123456"

Swap: Basic Usage



The screenshot shows a web browser window with the URL 192.168.56.101. The page has a dark purple header with the SWAP logo, navigation links for 'Exchanges' and 'Settings', and a user dropdown menu showing 'Administrator'. The main content area is titled 'Enrollments summary' and includes a blue 'Export enrollments' button. Below this is a table with the following data:

First year		
Semester	Course	Enrollments
1st	Álgebra Linear EI	0
1st	Cálculo	0
1st	Elementos de Engenharia de Sistemas	0
1st	Laboratórios de Informática I	0
1st	Programação Funcional	0
1st	Tópicos de Matemática Discreta	0
2nd	Análise	0
2nd	Laboratórios de Informática II	0
2nd	Lógica EI	0

Swap: Advanced Deployment

